

Windows Update Repair

Covers: Update scan failures, "Can't connect to update service," endless "Please wait," stuck downloads, update loop after reboot, post-feature-upgrade hangs.

Decision Tree

```
Windows Update not working
■■■ Shows "Please wait" indefinitely
■   ■■■ Recently upgraded (feature update)? → SCENARIO C
■   ■■■ No recent upgrade → SCENARIO A
■■■ "Can't connect to the update service"
■   ■■■ Can you reach the internet? (ping 8.8.8.8)
■   ■   ■■■ NO → Fix network first
■   ■   ■■■ YES → Can you resolve DNS? (nslookup update.microsoft.com)
■   ■       ■■■ NO → Flush DNS, check DNS server config
■   ■       ■■■ YES → WSUS residue or proxy issue → SCENARIO B
■■■ Update downloads but fails, retries every reboot → SCENARIO D
■■■ Specific error code displayed → Search the code before running generic fixes
```

SCENARIO A: Standard Reset

Orchestrator:

```
.\Invoke-HelpDeskOrchestrator.ps1 -RepairWindowsUpdate
```

Manual:

```
Stop-Service wuauserv, bits, cryptsvc, msiserver -Force
Rename-Item "C:\Windows\SoftwareDistribution" "SoftwareDistribution.old" -Force
Rename-Item "C:\Windows\System32\catroot2" "catroot2.old" -Force
Start-Service cryptsvc, bits, wuauserv, msiserver
Usoclient.exe StartScan
```

Check after 2-3 minutes. Still broken → Scenario B.

SCENARIO B: WSUS Residue / Proxy

Machines previously managed by WSUS often retain registry keys pointing to a dead server.

Orchestrator:

```
.\Invoke-HelpDeskOrchestrator.ps1 -RepairWindowsUpdate -ClearLegacyWsusPolicy -ResetWindowsUpdatewin
```

Manual:

```
Remove-ItemProperty "HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate" -Name "WUService" -ErrorAction SilentlyContinue
Remove-ItemProperty "HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate" -Name "WUStatusServer" -ErrorAction SilentlyContinue
Remove-ItemProperty "HKLM:\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate\AU" -Name "UseWUService" -ErrorAction SilentlyContinue
netsh winhttp reset proxy
Restart-Service wuauclt
UsbClient.exe StartScan
```

SCENARIO C: Post-Feature-Update Hang

1. Run Scenario A first.
2. Check for pending reboot:

```
Get-ItemProperty "HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\Component Based Servicing" -Name "RebootPending"
```

3. If pending, reboot first, then retry.
4. Still stuck → Run Windows Update troubleshooter from Settings.

SCENARIO D: Failed Update Retries Every Reboot

1. Identify the failing update:

```
Get-WindowsUpdateLog
```

2. Hide the problematic update:

```
Install-Module PSWindowsUpdate -Force
Hide-WindowsUpdate -KBArticleID "KB5XXXXXX" -Confirm:$false
```

3. If it's a required security update, download manually from catalog.update.microsoft.com and install:

```
wusa.exe C:\path\to\update.msu /quiet /norestart
```

Escalation

If Scenarios A-D fail:

- `DISM /Online /Cleanup-Image /RestoreHealth`
- `sfc /scannow`
- In-place upgrade (mount ISO, run setup.exe, select "Keep everything")
- Reimage as last resort